

Circuit Analysis and Design

EE-110L

Laboratory Manual



Student Name: _____

Registration No. _____

Lab Section: _____

Department of Electrical Engineering

University of Engineering and Technology, Lahore



LABORATORY SAFETY RULES

(Safety Committee, Electrical Department UET Lahore)

The following general rules and safeguards should be followed always in the EED laboratories.

Administrative

1. Do not block access to these electrical panels/boards and shut-off switches.
2. Smoking is prohibited.
3. No food or drinks allowed.
4. Do not run or engage in reckless behavior.
5. Wear Personal Protective Equipment (PPE) if required.
6. Workspace should be clear of unnecessary material such as extra books, paper, purses, and clothes.
7. Never wear rings, watches, bracelets, necklaces, or other electrically conductive jewelry.
8. No chatting or gaming is allowed.
9. Inappropriate use of Internet (chatting, gaming and pornography etc.) is strictly prohibited.

Electrical safety

1. Working alone and unsupervised is forbidden unless getting permission from HSE incharge, security incharge, and safety officer.
2. When checking an operating circuit, keep one hand either in a pocket or behind your back to prevent current from passing through your chest cavity and injuring your heart.
3. Be familiar with the electrical hazards associated with your workplace.
4. Avoid contacting circuits with wet hands or wet materials.
5. Use electrical cords only if they are in good condition.
6. Avoid being grounded. Stay at least 6 inches away from all metal materials, walls, and water sources.
7. When unplugging a power cord, pull on the plug, not on the cable.
8. Follow lockout-tag out safety procedure.
9. Equipment found to be faulty in any way should be reported to the lab incharge immediately.
10. When attaching a high voltage power supply, always switch off the supply.
11. When disassembling a circuit, first remove the source of power.

Emergency Response

1. Be familiar with emergency phones, fire alarm posters, emergency exit routes, first aid boxes, evacuation plans, eye wash, assembly points, and location of fire extinguishers.
2. Inform your lab instructor immediately after any incident, injury, fire or explosion.

Common Sense

1. It is better not to touch anything with which you are not familiar.
2. Your personal laboratory safety depends mostly on you.



Safety Documents



ELECTRICAL SAFETY RULES FOR ELECTRICAL ENGINEERING LABORATORIES

(Safety Committee, Electrical Department UET Lahore)

1. For experiments where the voltages are 50 V AC (RMS)/DC or higher there should be at least two students in the laboratory and they must be supervised by either an instructor or a laboratory staff.
2. Switch off power by operating an associated switch or a circuit breaker before working on a circuit. It is a good practice to unplug the power cord from the source.
3. Complete your wiring, think about it, discuss it with your partners and re-check before switching power on. If unsure, ask an instructor or a laboratory staff.
4. Any alteration/modification in the circuit must only be done after the power to the circuit has been switched off (point 2).
5. When working with energized circuits, always use one hand: the other hand must be at your back. Think about this: it is absolutely important for your safety. It will save from possible electrocution.
6. After completing an experiment, switch off the power before dismantling the circuit.
7. Switch off power before checking or replacing a fuse. Identify and correct the cause of a blown-up fuse or a tripped circuit breaker before replacing the fuse or re-setting the circuit breaker.
8. Do not use damaged cords/leads, cords/leads which become hot or cords/leads with exposed wiring. Report to the laboratory staff if this happens.
9. If measurements must be made on live or energized circuits, use well-insulated meter probes. Remember: work with only one hand!
10. Use extension cords when necessary, and only on a temporary basis. Do not join leads together to make a longer lead/connection in a circuit.
11. Do not come to a laboratory wearing a chappal or slippers. The students working with electrical machines or with other rotating parts should not wear loose clothes.
12. Do not bring and consume edible items and drinks/beverages in a laboratory.
13. Avoid working with wet hands and clothing.
14. You should remove loose metallic bangles, bracelets, necklaces, earrings and watchstraps before working on an electrical circuit. You should not have long loose hair as well!
15. Always check the electrical ratings of the equipment you work on and make sure you operate it within its ratings.
16. Never over-load an electrical circuit.
17. The fuses and circuit breakers must never be bypassed. Never replace a low-current fuse with a higher current rating fuse.
18. Make sure chassis or cabinets are grounded.
19. Safely discharge capacitors in equipment before working on a circuit.

Electrical Emergency Response

- It is the duty of a laboratory director, instructor and laboratory staff to make the students aware of the available Emergency Power-Off arrangement in their laboratory, and when and how they should operate it.

Dr. Muhammad Asghar Saqib
Convener, Department's Safety Committee



Department of Electrical Engineering,
University of Engineering and Technology, Lahore.

LABORATORY SAFETY RULES
(Safety Committee, Electrical Department UET Lahore)

Administrative



Do Not
Block Passage



No Jewellery



No Smoking



No Chatting &
Gaming



No Food or
Drink



No Unethical
Websites



No Running



Use Personal Protective
Equipment (PPE)



Keep your
Surroundings Clean

Electrical Safety



No Alone
Working



Work With
One Hand



Electrical
Hazards



No touch with
wet hand



Use good
cords



Not to be
Grounded



Pull the Plug
Out from Head



Do Not Pull the
Plug Out from Wire



Follow Lockout-
Tag Out

Emergency Response



Emergency Phones
and Cell Numbers



Emergency Exit



Know the
Evacuation plans



Fire
Extinguishers



Assembly Points



First Aid Box

Common Sense



Keep Calm and Use
Common Sense



Safety Starts with
You



Death of
Common Sense



Safety Documents

Issued by: HSE Incharge

Approved By: Chairman

Circuit Analysis & Design

LAB PERFORMANCE EVALUATION RUBRIC					
SUBJECT TITLE: EE-110L Circuit Analysis & Design (Lab)					
PERFORMANCE	CLOs Mapping	Excellent (100%)	Good (80%)	Satisfactory (60%)	Needs Improvement (0% - 40%)
Connections (5-Points)	CL01 CL02 CL03	Done all circuit connections properly by following the instructions given in the lab manual, in first attempt.	Done circuit connections, by following the instructions given in the lab manual, and debugging the circuit on its own.	Done circuit connections, by following the instructions given in the lab manual, and could not debug the circuit without assistance of lab instructor.	Could not complete circuit connections by following the instructions given in the lab manual.
Execution of Lab & Pre-lab Tasks (20-Points)	CL01 CL02	Properly done the experiment and results shown to the instructor. Used time well in lab and focused attention on the experiment. Routinely provides useful ideas when participating in the group and in classroom discussion. Actively looks for and suggests solutions to problems.	Experiment results had minor issues but were corrected on its own. Used time well. Stayed focused on the experiment most of the time. Usually provides useful ideas when participating in the group and in classroom discussion.	Experiment results had issues and could not correct own its own. Did the lab but did not appear very interested. Sometimes it provides useful ideas when participating in the group and in classroom discussion. A satisfactory group member who does what is required.	Did not try to perform the experiment, Participation was minimal, or student was hostile about participating. Rarely provides useful ideas when participating in the group and discussion. May refuse to participate.
Presentation of Experimental Data (5-Points)	CL01 CL02	Professional looking and accurate representation of the data in tables and/or graphs. Graphs and tables are labeled and titled. Conclusion includes whether the findings supported the hypothesis, possible sources of error, and what was learned from the experiment.	Accurate representation of the data in tables and/or graphs. Graphs and tables are labeled and titled. Conclusion includes whether the findings supported the hypothesis and what was learned from the experiment.	Accurate representation of the data in written form, but graphs or tables are not properly drawn/presented. Conclusion includes what was learned from the experiment.	Data are not shown or are inaccurate. No conclusion was included in the report or shows little effort and reflection.
Simulation Task (25)	CL03	Properly used the modern tools/software to execute the task (by following the given instructions) to produce required outcomes and results shown to the instructor.	Properly used the modern tools/software to execute the task (by following the given instructions). Results had minor issues but corrected own its own.	Tried to follow the instructions to use the modern tools/software for the execution of the task. Simulation results had issues and could not correct these issues on its own.	Did not try to use modern tools/software to perform the task. Showed a little or no interest in the use of modern tools/software.

Circuit Analysis & Design

Discipline and Safety (5-points)	CLO4	Came in the lab in-time with full preparation, used time well and focused attention on the experiment. Stayed in seat and got up for a specific lab-related reason, took care of lab-related business and sat down right away. Kept the working area clean and in proper order. Lab is carried out with full attention to relevant safety procedures & directions.	Came late (within 5 min) with full preparation, used time well and stayed focused on the experiment most of the time. Stayed in seat and got up for a specific lab-related reason. Good job on cleaning up working area and handling the equipment. Lab is generally carried out with attention to relevant safety procedures & directions.	Came late (more than 5 min), not properly prepared, did the lab but did not appear very interested. A satisfactory student who does what is required. Got out of seat and wandered around. Had to be reminded to clean up working area and careless about the equipment use. Lab is carried out with some attention to relevant safety procedures & directions.	Came late (more than 15 min), did not prepare at all, did not look interested in lab work, did not try to focus the attention on lab task. Got out of seat and wandered around. Chased others, ran, or played around. Had to be asked by teacher to keep work area clean, did not clean up area or refused to clean up. Safety procedures were ignored and did not follow directions.
TOTAL	<-----Total points earned (Max 35) = Lab Performance Grade ----->				

Experiment No. 1

Study of the Transient and Steady-State Response of Series RL and RC Circuits

CLO Mapping:

CLOs	Description	Domain	Taxonomy Levels	PLOs Levels
CLO1	Demonstrate the basic concepts related to RC, RL and RLC circuits.	Psycho-motor	Level 3 Precision	GA-1 Engg. Knowledge
CLO4	Practice appropriate professional and ethical standards during the lab.	Affective	Level 2 Responding	GA-8 Engg. Ethics

Objectives:

At the end of this lab session students will be able to:

- Apply a pulse waveform to series RL and RC circuits.
- Observe and analyse the transient response of the circuits.
- Study the effect of pulse width relative to the circuit time constant.
- Compare the behaviour of the RL and RC circuits under pulse excitation.

Pre-requisite (Theory):

- Article 7-1, 7-2 from the Textbook

Apparatus:

- Oscilloscope
- Signal generator
- Inductors
- Capacitor
- Resistors

Theoretical Background (RL Series Circuit)

At the instant when step voltage is applied to an RL network, the current increases gradually and takes some time to reach the final value. The reason current does not build up instantly to its final value is that as the current increases, the self-induced e.m.f. in inductor opposes the change in current (Lenz's Law). Mathematically,

$$i(t) = \frac{V}{R}(1 - e^{-t/\tau})$$

Where t = time elapsed since pulse is applied and $\tau = L/R$ = Time constant of the circuit. During the next half cycle of pulse, when the pulse amplitude is zero, the current decreases to zero exponentially. Mathematically,

$$i(t) = \frac{V}{R}e^{-t/\tau}$$

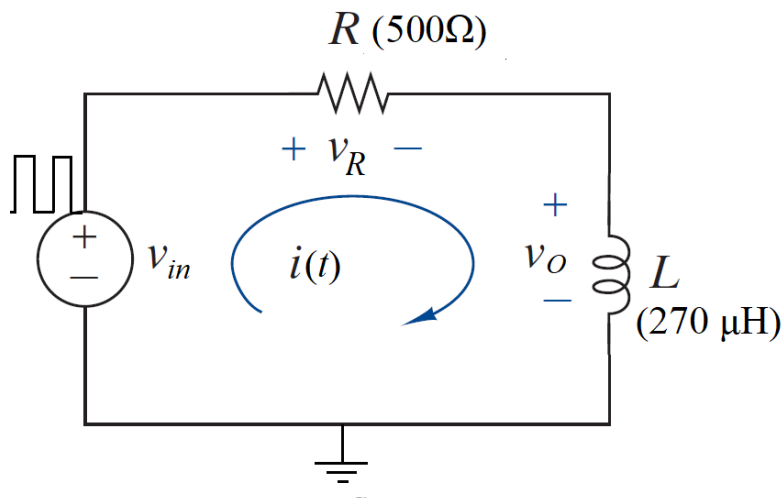


Figure 1: A series RL circuit

Task1:

1. Set the output of the function generator to a square-wave with peak-to-peak amplitude of 5V and adjust the frequency that should allow the inductor to charge and discharge fully i.e. f should be less than or equal to $f_{max} = \frac{1}{10\tau}$. Use dc offset to obtain pulse train ranging from 0 to 5 volts from square wave.
2. Patch the circuit given in Fig. 1 on the breadboard and apply pulse train to it.
3. Display simultaneously voltage $v_{in}(t)$ across the function generator (on CH 1) and $v_o(t)$ across the inductor L (on CH 2). Ensure common ground for both channels and the signal generator.
4. Sketch the two measured waveforms $v_{in}(t)$ and $v_o(t)$, calculate and sketch the waveforms, $v_R(t)$ and $i(t)$. Label the time, voltage and current scales. Note that the voltage waveform $v_R(t)$ across resistor R, also represents the current $i(t)$ waveform.
5. Measure the time constant, τ , using the waveform $v_o(t)$. Expand the time scale and measure the time it takes for the waveform to complete **63%** of its total change, i.e. 5V. Note the measured value of τ .
6. Compare values of theoretically expected and experimentally obtained time constants τ .
7. Now calculate and measure $i(t)$ for five different values of ' τ ' for rise of current as well as fall of current.

Observations & Calculations:

Write down the values of following parameters:

$$R = \underline{\hspace{2cm}}, \quad L = \underline{\hspace{2cm}}, \quad \tau = \underline{\hspace{2cm}},$$

$$10 \tau = \underline{\hspace{2cm}}, \quad f_{max} = \underline{\hspace{2cm}}$$

Rise of Current

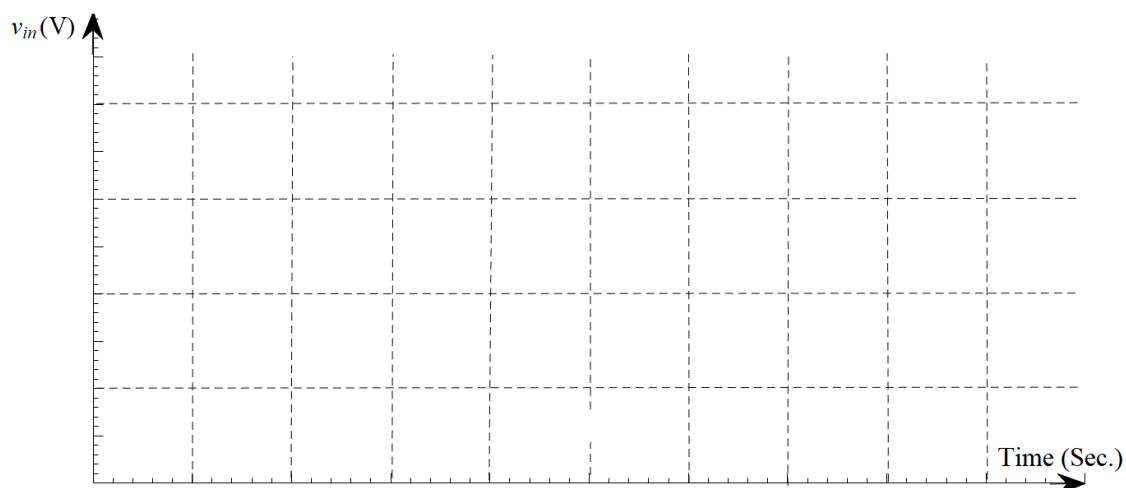
Calculate current using $i(t) = \frac{V}{R}(1 - e^{-t/\tau})$ and measure current (from oscilloscope)

No. of Time Constants	Calculated Current (amp)	Measured Current (amp)
τ		
2τ		
3τ		
4τ		
5τ		

Fall/Decay of Current

Calculate current using $i(t) = \frac{V}{R}(1 - e^{-t/\tau})$ and measure current (from oscilloscope)

No. of Time Constants	Calculated Current (amp)	Measured Current (amp)
τ		
2τ		
3τ		
4τ		
5τ		



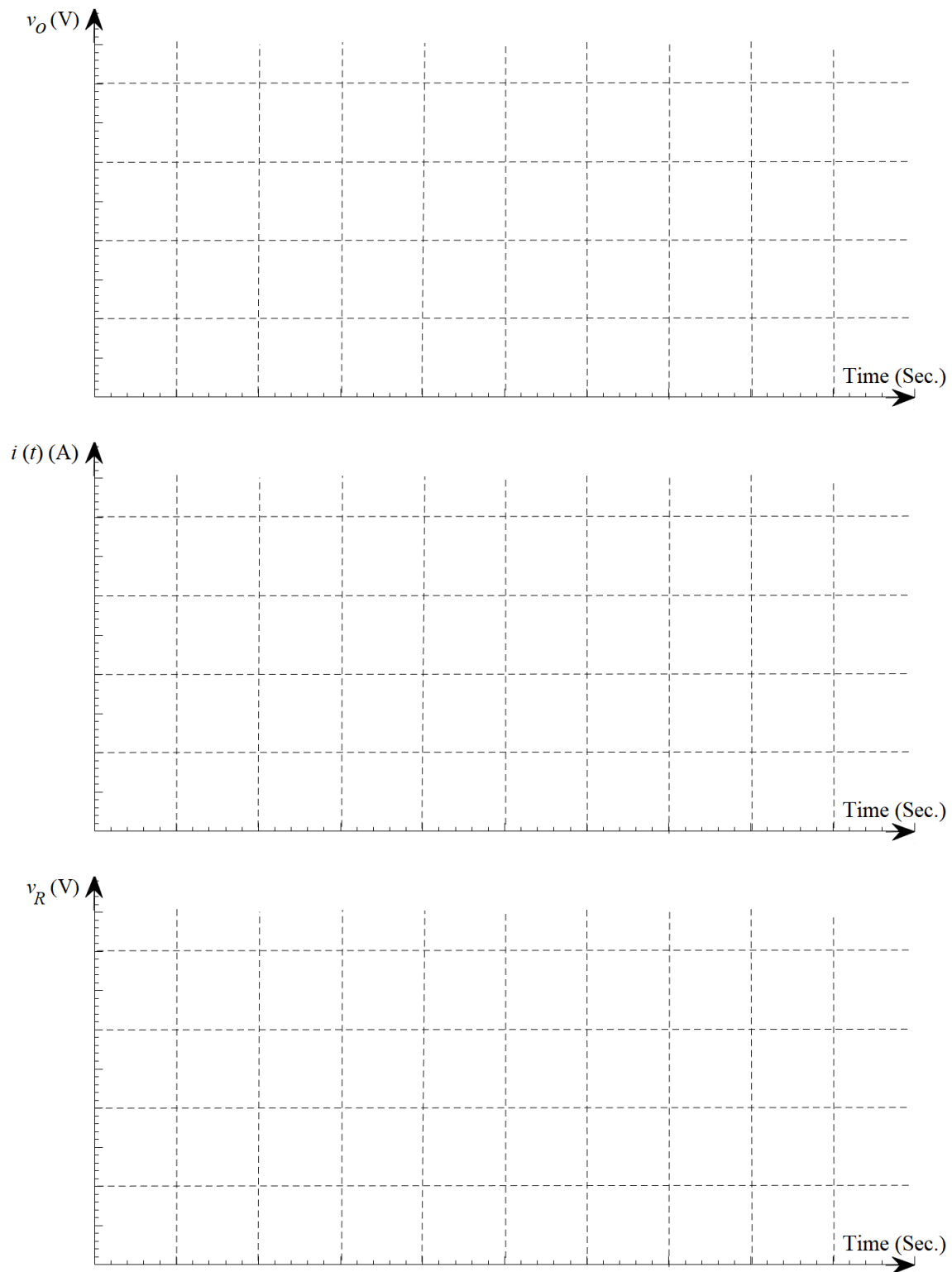


Figure 2: Input voltage, output voltage, loop current and resistor voltage of a series RL circuit.

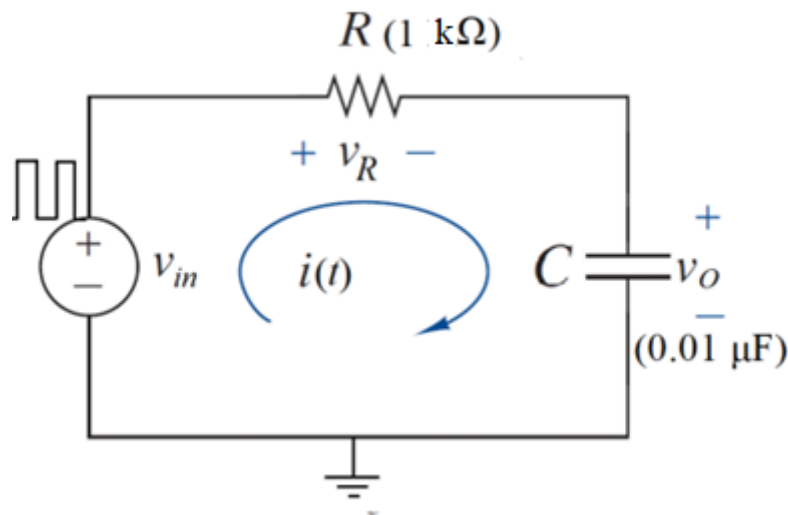


Figure 3: A Series RC circuit

Theoretical Background (RC Series Circuit)

A capacitor can be charged by connecting its two terminals to the two terminals of a battery. If the capacitor is then disconnected from the battery, it will retain this charge and the potential across the capacitor will remain that of the battery. If a resistor is then connected across the capacitor, charge will flow through the resistor until the potential difference between the two terminals goes to zero. The potential will decrease with time according to the relation:

$$v(t) = v_0 e^{\frac{-t}{\tau}}$$

where v_0 represents the voltage at time $t = 0$, and ‘ τ ’ represents the “time constant” or time that it takes for the voltage to decrease by a factor of $1/e$.

If the frequency of the square wave v_{in} is too high (i.e. if $f \gg 1/RC$), then v_0 and v_R will not have enough time to reach their asymptotic values. If the frequency is too low (i.e. if $f \ll 1/RC$), the decay time will be very short relative to the period of the waveform and thus the exponential decay will be difficult to observe. As a rough guideline, the period of the square wave should be chosen such that it is approximately equal to $10RC$, for the responses to be readily observed on an oscilloscope.

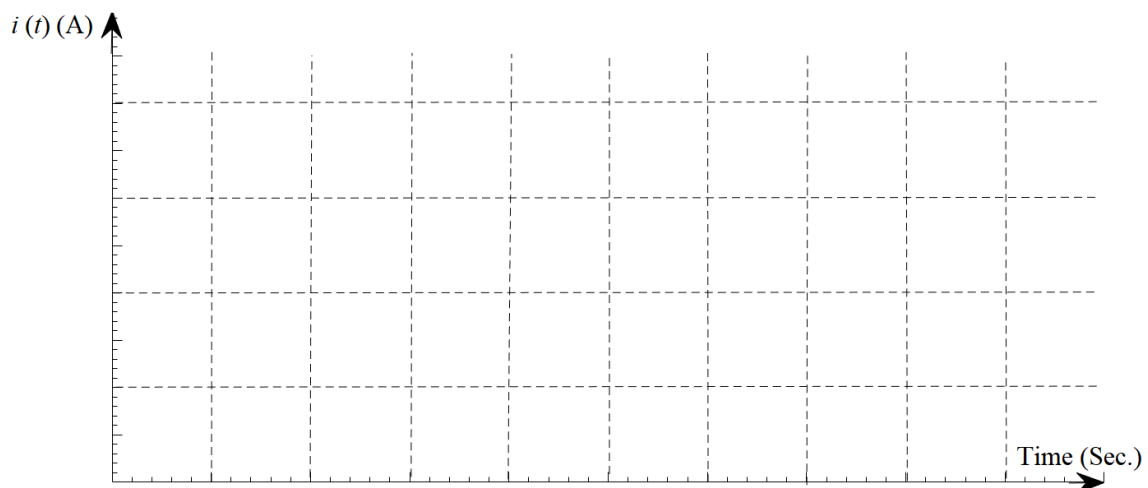
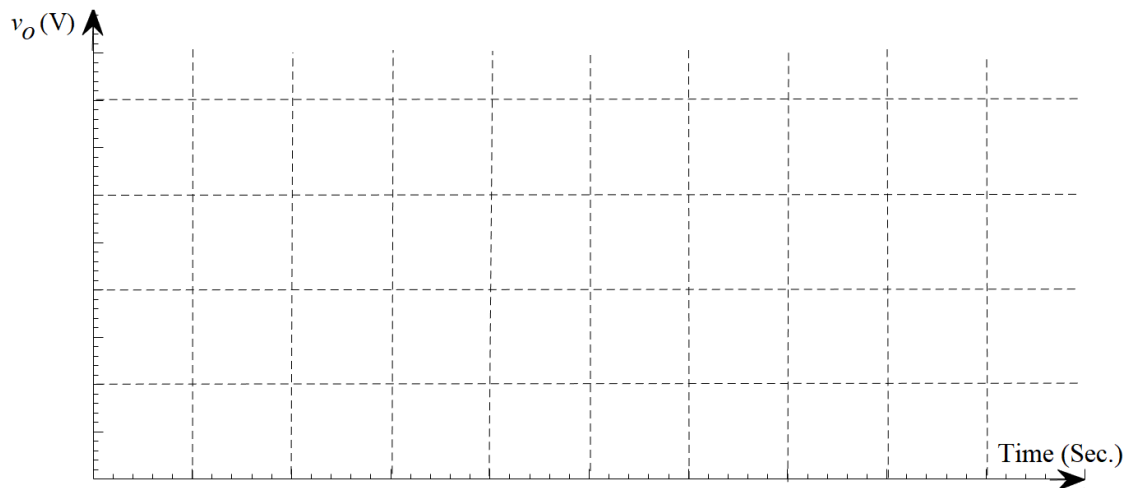
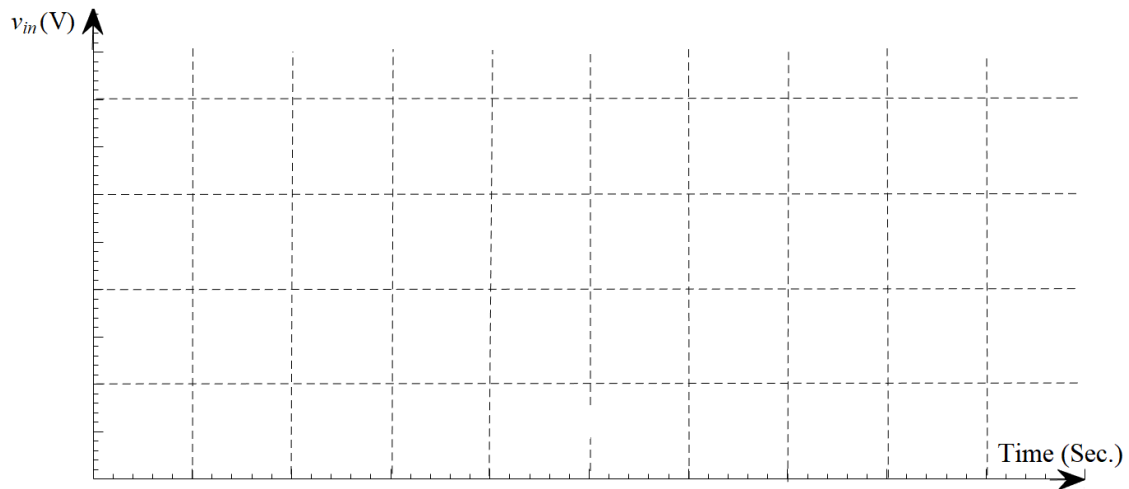
Task2:

1. Set the output of the function generator to a square-wave with peak-to-peak amplitude of 5V and adjust the frequency that should allow the capacitor to charge and discharge fully i.e. f should be less than or equal to $f_{max} = \frac{1}{10\tau}$. Use dc offset to obtain pulse train ranging from 0 to 5 volts from square wave.
2. Patch the circuit given in **Figure. 3** on the breadboard and apply pulse train to it.
3. Display simultaneously voltage $v_{in}(t)$ across the function generator (on CH 1) and $v_o(t)$ across the capacitor C (on CH 2). Sketch the two measured waveforms $v_{in}(t)$ and $v_o(t)$, also calculate, sketch and label the waveforms, $v_R(t)$ and $i(t)$ on the graphs given in **Figure. 4**.

4. Measure the time constant, ' τ ' using the waveform $v_o(t)$. Expand the time scale and measure the time it takes for the waveform to complete 63% of its total change, i.e. 5V. Compare values of theoretically expected and experimentally obtained time constants ' τ '.

Calculated ' τ ' = _____,

Measured ' τ ' = _____



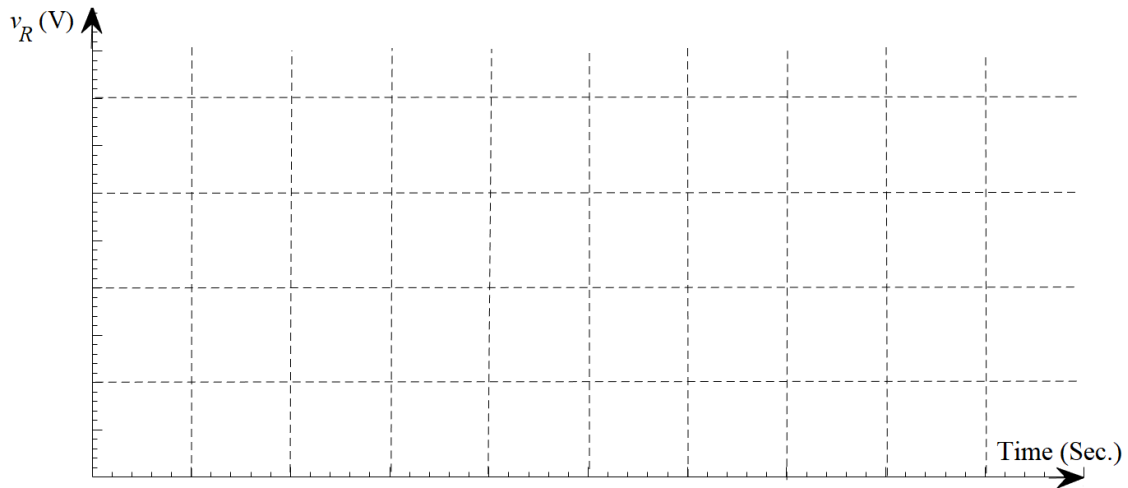


Figure 4: Input voltage, output voltage, loop current and resistor voltage of a series RC circuit.

Discussion:

Q1: What would be the effect on v_0 and v_R , if the resistance value is halved in Fig. 1?

Q2: What would be the effect on v_0 and v_R , if the inductor value halved in Fig. 1?

Q3: What would be the effect on v_0 and v_R , if the capacitance value is halved in Fig. 3?

Lab Instructor: _____

Dated: _____